

MLOps Foundations Solutions and Marking Guide

SRAI Book 5 Lesson 1 | PU-B05-C01 | Instructor Controlled v0.1

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This guide supplies expected reasoning and marking anchors. Equivalent technically sound answers receive credit when they preserve evidence, failure behavior and accountable authority.

Exercise 1

A complete map covers framing, accepted data, experiment, validation, registry, approval, deployment, monitoring, change and retirement. It names data owner, builder, evaluator, operator, domain owner and approver. Mandatory gates are conjunctive. Full marks require a sector-specific return path after drift or incident.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Map the lifecycle for a monthly public-sector prediction service. Identify states, artifacts, owners, mandatory gates, exception authority and return paths. Apply it to one of Agriculture, Education, Health or Habitat.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 12 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 2

The contract identifies immutable or versioned inputs and the acceptable reconstruction tolerance. A digest proves byte identity under the selected algorithm. It does not prove correctness, provenance, fairness, authorization or semantic equivalence.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Write a contract covering data, code, configuration, environment, random state, hardware assumptions and tolerance. Explain what a cryptographic digest proves and what it cannot prove.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 12 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 3

Strong answers test exact schema, types, units, ranges, missingness, categories, uniqueness where relevant and time coverage. A failed mandatory test blocks downstream execution. An exception requires an owner, rationale, impact assessment and expiry.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Design schema, range, missingness, temporal and category checks for the selected sector. Inject one failure in the notebook and show that training stops.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 12 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 4

Precision is $160/240 = 0.667$. Recall is $160/200 = 0.800$. Specificity is $220/300 = 0.733$. Accuracy is $380/500 = 0.760$. F1 is approximately 0.727. Threshold choice must connect false negatives, false positives and intervention capacity.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Using a confusion matrix with TP 160, FP 80, FN 40 and TN 220, calculate precision, recall, specificity, accuracy and F1. Recommend a threshold policy linked to consequences and intervention capacity.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 16 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 5

The manifest links exact data, split, code, configuration, environment, model digest, metrics and decision record. Registration stores a candidate. Eligibility means declared technical gates passed. Approval is an authorized decision. Deployment changes the serving state.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Create a manifest that binds dataset, split, code, environment, model, evaluation and decision records. Distinguish registration, eligibility for review, approval and deployment.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 12 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 6

The contract defines required fields, units, values, missing behavior, response schema, model identity and errors. Monitoring includes latency, errors, data quality, distributions, outcomes, slice performance, security and cost, each with threshold, owner and runbook.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Specify an inference contract and a monitoring plan covering service, data, prediction, outcome, subgroup, security and cost indicators. Assign owners and responses.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 14 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 7

Containment rejects or quarantines malformed input and may roll back to a known-good manifest. Impact review identifies affected predictions and decisions. Recovery restores the complete contract. The incident record documents detection, containment, decisions, verification and lessons.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Respond to a silent unit change in a mandatory input. Define detection, containment, impact assessment, rollback, recovery verification and post-incident evidence.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 10 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Exercise 8

Technical evidence cannot create institutional authority. Remediation means completing missing evidence and obtaining an authenticated named decision. Approval has scope, conditions and review date. Retirement revokes traffic and credentials, archives required evidence and confirms downstream closure.

Criterion	Share	Full-credit evidence
Technical correctness	40%	Explain why a technically ready model remains blocked without approval. Define authorized remediation, review expiry and retirement evidence.
Applied reasoning	30%	Links the control to consequence and failure behavior
Evidence specificity	20%	Names artifacts, versions, owners and decision records
Clarity	10%	Separates fact, inference, recommendation and authority

Maximum: 12 marks. Partial credit should reflect the four criteria; do not award full credit for terminology without an operational decision path.

Canonical notebook checks

Control	Expected
Seed	42
Notebook	V5_N01_mlops_foundations.ipynb
Code cells	22 executed without error
Data	Synthetic agricultural-risk records
Manifest	Valid JSON with native Python scalar types
Authorization	False during canonical run
Decision	Blocked pending named governance approval